TS1: Síntesis y operaciones de señales

Ejercicio 2

$h[n] = \delta[n] - \delta[n-4]$; ¿ $y[n] = x[n] * h[n]$?

$$y[n] = \sum_{i=-\infty}^{+\infty} x[i] \cdot \underbrace{h[n-i]}$$

↳ $h[n-i] = \delta[n-i] - \delta[n-i-4]$

↓

La delta funciona como un selector, en este caso me va a seleccionar dos valores para cada n

→ Para cada valor de n voy a tener solo dos valores de h

↳ $n - i = 0 \rightarrow n = i$

$n - i - 4 = 0 \rightarrow n - 4 = i$

a. $x[n] = \cos(w_0 . n . T_s)$

- $n=0$: $y[0] = \underbrace{x[0] \cdot h[0]}_{i=0} + \underbrace{x[-4] \cdot h[0]}_{i=-4}$

  → $y[0] = 1 - \cos(4w_0 . T_s)$

- $n = 1$: $y[1] = x[1] . h[0] + x[-3] . h[0]$

  → $y[1] = \cos(w_0 . T_s) - \cos(3w_0 T_s)$

- $n = 2$: $y[2] = x[2] . h[0] + x[-2] . h[0]$

  → $y[2] = \cos(2w_0 . T_s) - \cos(2w_0 T_s) = 0$

- $n = 3$: $y[3] = x[3] . h[0] + x[-1] . h[0]$

  → $y[3] = \cos(3w_0 . T_s) - \cos(w_0 . T_s)$

- $n = 4$ $y[4] = x[4] . h[0] + x[0] . h[0]$

  → $y[4] = \cos(4w_0 . T_s) - 1$

→ $y[n] = \cos(n w_0 . T_s) - \cos((n-4) . w_0 . T_s)$

→ Para expresar el resultado como un único coseno de la forma $A . \cos(w_0 . n . T_s + \varphi)$ usamos la siguiente propiedad trigonométrica...

$$\cos(A) - \cos(B) = 2 . \operatorname{sen}\left(\frac{B-A}{2}\right) . \operatorname{sen}\left(\frac{A+B}{2}\right)$$

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→ En nuestro caso tenemos que ...

$$\left.\begin{array}{l} A = \omega_0 . n . T_s \\ B = (n-4) . \omega_0 T_s \end{array}\right\} \rightarrow \frac{B-A}{2} = -2\omega_0 . T_s \quad ; \quad \frac{A+B}{2} = (n-2)\omega_0 T_s$$

→ $y[n] = 2 . \text{sen}(-2\omega_0 T_s) . \text{sen}((n-2)\omega_0 . T_s)$

$y[n] = -2 . \text{sen}(2\omega_0 . T_s) . \text{sen}((n-2)\omega_0 . T_s)$

$\text{sen}(-x) = -\text{sen}(x)$

→ Ahora uso la siguiente propiedad ... $\text{sen}(x) = \cos(x - \pi/2)$

→ $y[n] = -2 . \text{sen}(2\omega_0 T_s) . \cos(\underbrace{(n-2)\omega_0 . T_s - \pi/2}_{(n-2)\omega_0 . T_s - \pi/2 \,=\, n\omega_0 . T_s - 2\omega_0 T_s - \pi/2})$

$y[n] = -2 . \text{sen}(2\omega_0 . T_s) . \cos(n\omega_0 T_s - 2\omega_0 T_s - \pi/2)$

$$\boxed{\begin{array}{l} y[n] = A . \cos(\omega_0 n T_s + \varphi) \\ \text{... donde} \quad A = -2 . \text{sen}(2\omega_0 . T_s) \quad , \quad \varphi = -2\omega_0 T_s - \pi/2 \end{array}}$$

b- $x[n] = (1/2)^n . u[n]$

→ $u[n] = \begin{cases} 1 & , \; n \geq 0 \\ 0 & , \; n < 0 \end{cases}$

→ Voy a tomar solo los $i \geq 0$, porque para los $i < 0$ la función $x[i] = 0$

- $n = 0$: $y[0] = x[0] . h[0]$

  → $y[0] = 1$

- $n = 1$: $y[1] = x[1] . h[0]$

  → $y[1] = 1/2$

- $n = 2$: $y[2] = x[2] . h[0]$ → $y[2] = 1/4$
- $n = 3$: $y[3] = x[3] . h[0]$ → $y[3] = 1/8$
- $n = 4$: $y[4] = x[4] . h[0] + x[0] . h[0]$ → $y[4] = -15/16$

→ $y[n] = x[n] - x[n-4]$

$$\boxed{y[n] = \left(\frac{1}{2}\right)^n . u[n] - \left(\frac{1}{2}\right)^{n-4} . u[n-4]}$$

$y[n] = \left(\frac{1}{2}\right)^n . (u[n] - 16 . u[n-4])$

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c. $x[n] = u[n+1] - u[n-2]$

→ $u[n+1] = \begin{cases} 1 , & n \geq -1 \\ 0 , & n < -1 \end{cases}$

→ $u[n-2] = \begin{cases} 1 , & n \geq 2 \\ 0 , & n < 2 \end{cases}$

- Para $n \leq 2$ → $x[n] = 0$
- Para $-1 \leq n \leq 1$ → $x[n] = 1 - 0 = 1$
- Para $n \geq 2$ → $x[n] = 1 - 1 = 0$

→ $x[n] = \begin{cases} 1 , & -1 \leq n \leq 1 \\ 0 , & \text{cualquier otro caso} \end{cases}$

→ Voy a tomar los $i \in [-1; 1]$, porque para cualquier otro valor de i, $x[i] = 0$

- $n = 0$: $y[0] = x[0] . h[0]$ → $y[0] = 1$
- $n = 1$: $y[1] = x[1] . h[0]$ → $y[1] = 1$
- $n = 2$: $y[2] = 0$
- $n = 3$: $y[3] = x[-1] . h[0]$ → $y[3] = -1$
- $n = 4$: $y[4] = x[0] . h[0]$ → $y[4] = -1$
- $n = 5$: $y[5] = x[1] . h[0]$ → $y[5] = -1$

→ A partir de ahí, para todos los n que siguen, la salida $y[n] = 0$

→ $y[n] = x[n] - x[n-4]$

$$\boxed{y[n] = u[n+1] - u[n-2] - u[n-3] + u[n-6]}$$

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